

The Integer 31 as a Discrete Geometric Node in Kochenov G

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Abstract

The integer 31 is found to be the optimal integer satisfying two independent geometric constraints within the Kochenov parametrization of the gravitational constant. (1) Quadratic closure: $31^2 = 961 = G_{\text{Koch}} \times 10^{11} \times 144$ with a gap of 0.001%. (2) Proximity to the continuous optimum $n \approx 31.338$ defined by $(\phi^6 \times n)^4 = 10^{11}$. No neighbouring integers (30 or 32) satisfy both constraints simultaneously. The integer 31 provides the nearest discrete anchor.

1 Spiral scaling step

Define the structural block

$$S = \phi^6 \times 31 \approx 556.272429.$$

The logarithmic scaling step is

$$\kappa = \log_{10}(S) \approx 2.745288.$$

Four steps give

$$4\kappa \approx 10.981150.$$

The target value is $\log_{10}(10^{11}) = 11.000000$. The gap is

$$11.000000 - 10.981150 = 0.018850 \quad (\approx 0.171\% \text{ of } 11).$$

A correction term $12/720 = 0.016667$ reduces the gap. After adding it:

$$10.981150 + 0.016667 = 10.997817,$$

leaving a residual

$$11.000000 - 10.997817 = 0.002183 \quad (\approx 0.020\%).$$

This residual is a numerical remainder; the expression is an approximation, not an exact identity.

2 Two geometric constraints on 31

Constraint A – Quadratic closure:

$$31^2 = 961 = G_{\text{Koch}} \times 10^{11} \times 144,$$

with $G_{\text{Koch}} = \frac{31^2 \sqrt{5}}{\phi^{12} \times 10^{11}}$ and $144 = F_{12}$. The relative gap from the exact equality is 0.001% (within the Kochenov parametrization).

Constraint B – Scaling optimum:

$$(\phi^6 \times n)^4 = 10^{11}$$

yields $n \approx 31.338$. The integer 31 is the closest integer. The deviation is 1.08%; the role of the $1/720$ term is discussed separately in the full geometric identity.

3 Comparison with neighbouring integers

Replacing 31 by 30 or 32 worsens both constraints significantly:

n	Gap from CODATA (G)	Gap of $(\phi^6 n)^4$ from 10^{11}	Note
30	6.358%	16.018%	too small
31	0.011%	4.247%	optimal discrete node
32	6.544%	8.718%	too large
31.338	$\sim 0\%$	$\sim 0\%$	continuous optimum

4 Additional connection

$$31^2 = 961 = 367 \times \phi^2 \quad (\text{gap } 0.019\%).$$

This allows G to be written without an explicit 31^2 :

$$G_{\text{Koch}} = \frac{367 \times \sqrt{5}}{\phi^{10} \times 10^{11}} \quad (\text{gap } 0.019\%).$$

5 What this is not

The role of 31 is presented as an empirical observation. No claim is made that 31 is derived from first principles. The equality $(\phi^6 \times n)^4 = 10^{11}$ is not exact (the gap is 4.25% for $n = 31$). The continuous optimum $n \approx 31.338$ is not an integer. The integer 31 provides the best discrete approximation within the system.

6 Conclusion

The integer 31 satisfies two independent geometric constraints with the stated numerical residuals. No neighbouring integer achieves this combination. Within the Kochenov parametrization, 31 serves as a discrete anchor linking the golden ratio, the gravitational constant, and the integer 367.

7 Reference value

$$G_{\text{Koch}} = \frac{31^2 \times \sqrt{5}}{\phi^{12} \times 10^{11}} = 6.6735467449616181237591997717904236 \dots \times 10^{-11} \text{ m}^3\text{kg}^{-1}\text{s}^{-2}.$$

See also

- $\phi^{12}/\sqrt{5} = 144 + 1/720$ (Zenodo, April 2026)
<https://zenodo.org/records/19769041>
- Kochenov Geometric Identity for G (Zenodo, April 2026)
<https://zenodo.org/records/20009102>
- Geometric Origin of 10^{11} in the Gravitational Constant (Zenodo, April 2026)
<https://zenodo.org/records/20009474>

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